Sample title

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Basic concepts

- A *matroid* is a finite ground set E with a family $\mathcal{I}$ of independent subsets. The empty set is independent, subsets of independent sets are independent, and if $I, J \in \mathcal{I}$ with $|I| < |J|$, some $j \in J \setminus I$ makes $I \cup \{j\}$ independent.
- A canonical example is a *vector matroid*. Its elements label the columns of a matrix, and its independent sets label sets of linearly independent columns.
- Some matroids have no matrix representation. A matroid that can be represented by a matrix over a field $\mathbb{F}$ is called $\mathbb{F}$-representable.
- We need a few more concepts to state the problem. We omit detailed proofs due to time constraints.

- Analogous concepts from linear algebra still carry over.
  - A basis is a maximal independent set.
  - All bases have the same size, called the rank.
- We often describe matroids using a labelled ground set and a set of bases, e.g. $E = \{0, 1, 2\}$, $\mathcal{B} = \{\{1, 2\}, \{0, 2\}\}$.
- The dual M^* has the same ground set as M and bases $\{E \setminus B : B \in \mathcal{B}(M)\}$.
- Two fundamental operations give minors.
  - Deletion $M \setminus T$ has ground set $E \setminus T$. Its independent sets are the independent sets of M contained in $E \setminus T$.
  - Contraction by $T \subseteq E$ is $M/T = (M^* \setminus T)^*$.

Problem statement

- We classified all rank-4, 11-element excluded minors for $\mathbb{F}_5$-representability up to isomorphism.
- We call an $\mathbb{F}_5$-representable matroid a *quinary* matroid.
- An *excluded minor for $\mathbb{F}_5$-representability* is a nonquinary matroid M such that both M/e and $M \setminus e$ are quinary for every $e \in E(M)$.
- These are the minimal nonquinary matroids under the minor relation.
- The study of excluded minors is extremely important in matroid theory.

Comparison of methodology

- We used an LLM to solve the problem.
- Directly asking the question left the classification incomplete.
- We completed the classification by giving the LLM access to computational tools for an exhaustive search.

Proof outline

- The proof can be broken down into a few steps.
- The broad idea is to view an 11-element excluded minor M as an extension of a 10-element quinary matroid $N = M \setminus e$.
- For each N, we want extensions M which are nonquinary but all of whose deletions and contractions are quinary.
- First generate all appropriate N.
- Then enumerate all nonquinary M whose deletions are quinary. To do this, list the quinary extensions by e of N and of $N \setminus f$ for every $f \in E(N)$.
- Finally, pick those for which all single-element contractions are quinary. Then deduplicate up to isomorphism.

We shall give a bit more detail for each step, especially the computations involved.

Step 1

Generate 10-element quinary matroids. Due to a key result, we only need connected simple rank-4 ones up to isomorphism.
We say that two matrices over $\mathbb{F}_5$ are *field-isomorphic* if they differ by invertible row operations, nonzero column scalings and a column permutation.

- First generate the corresponding matrices up to field isomorphism. Start with I_4 by reducing a basis to the identity. By simplicity, it suffices to append unused projective points, one nonzero vector per one-dimensional subspace.
- Separately, obtain their labelled basis representations and deduplicate them up to matroid isomorphism.
- Sage provides methods to check field isomorphism and matroid isomorphism.

Step 2

Generate the candidate extensions for the contraction tests.
Let $P = \{0, \ldots, 9\}$ and take a new label $e = 10$. We work with labelled matroids, whose bases are subsets of a fixed ground set. Let $\mathcal{P}$ contain one labelled representative on P of each connected simple rank-4 quinary matroid on 10 elements.
Let $\mathcal{D}(N)$ be the set of simple rank-4 matroids M on $P \cup \{e\}$ with $M \setminus e = N$, with M nonquinary, and with $M \setminus a$ quinary for every $a \in P \cup \{e\}$.
We keep all labelled members of $\mathcal{D}(N)$. Our goal in this step is to exhaust $\mathcal{D}(N)$ for all $N \in \mathcal{P}$.
As mentioned before, we do this by enumerating extensions.

We first describe how to encode extensions.

- Let $N \in \mathcal{P}$ and suppose $M \setminus e = N$. For a rank-4 extension M, the unknown bases are of the form $T \cup \{e\}$ for triples $T \subseteq P$. Set $x_T = 1$ exactly when this set is a basis. An arbitrary x need not define a matroid.
- Let $\pi_f(x)$ restrict x to triples avoiding f. This gives the additional basis data for $M \setminus f$.
- Let $\mathcal{R}(Q)$ be the set of bit rows on triples of $E(Q)$ for simple rank-4 quinary extensions of Q by e.
  - $x \in \mathcal{R}(N)$ gives a quinary extension of N.
  - $\pi_f(x) \in \mathcal{R}(N \setminus f)$ gives a simple rank-4 quinary deletion at f.
- If all deletion rows are allowed, the full family defines a simple matroid. Any two proposed bases use at most eight labels, so basis exchange holds in a common deletion. Every pair of elements also survives in a simple deletion.
- The assignments corresponding exactly to $\mathcal{D}(N)$ are

$$\mathcal{S}(N) = \{x : \pi_f(x) \in \mathcal{R}(N \setminus f) \text{ for every } f \in P\} \setminus \mathcal{R}(N).$$

Step 2.1

For $Q = N$ or $Q = N \setminus f$, a labelled representation is a matrix over $\mathbb{F}_5$ with one column per element of Q. Its linearly independent column sets give the independent sets of Q. Two are labelled projectively equivalent if invertible row operations and nonzero column scalings relate them with the labels fixed.
We generate $\mathcal{L}(Q)$, one matrix per labelled projective equivalence class, as follows.

- For each field-isomorphism class in the catalogue whose matroid is isomorphic to Q, choose one matrix and relabel its columns using an isomorphism to Q. Then apply all automorphisms of Q to the column labels.
- Deduplicate the resulting matrices up to labelled projective equivalence by normalisation.

Step 2.1: computations

To compute $\mathrm{Aut}(Q)$, construct a coloured bipartite graph.

- Each element is a blue vertex and each basis is a red vertex.
- Connect a blue vertex to a red vertex exactly when the element belongs to the basis.
- The colour-preserving graph automorphisms, which Sage can compute, induce exactly $\mathrm{Aut}(Q)$ on the element vertices.

For normalisation, first fix an ordered basis B and use Gaussian elimination to make its columns I_r. Let the remaining columns form Z. The basis data of Q determine which Z_{ij} are nonzero. Blocks Z and Z' obtained from labelled projectively equivalent matrices can only differ by row and column scalings.

Consider a bipartite support graph.

- Each row of Z gives a vertex on one side and each column of Z gives a vertex on the other side.
- Connect row i to column j exactly when $Z_{ij} \neq 0$.
- Fix a spanning tree in each component.
- In each tree, fix the root multiplier at 1 and choose the remaining multipliers to make every tree entry 1. This gives a unique normalised block for the equivalence class.

Step 2.2

Build $\mathcal{R}(Q)$.
Let $V \in \mathcal{L}(Q)$ and let v represent an unused projective point. The matrix $[V \mid v]$ represents a simple rank-4 quinary extension of Q by e. Its row is

$$z_T = \mathbf{1}\{\det[\, V_T \mid v \,] \neq 0\}, \qquad T \in \binom{E(Q)}{3}.$$

The distinct rows over all such V and v are exactly $\mathcal{R}(Q)$.
For each V, we compute the determinant tests in one shot. Expand along v to write each determinant as $c_T^{\mathsf{T}} v$. Stack the cofactor rows in C and the projective point columns in D. The product CD over $\mathbb{F}_5$ gives all the tests at once.

Step 2.3

Compressing the representation.

- Right now, entries in $\mathcal{R}(Q)$ are recorded with a bit for each triple $T \subseteq E(Q)$, in lexicographic order.
- Dependent triples have bit 0. Independent triples spanning the same hyperplane have equal bits, so we can use one bit per hyperplane.
- Let $\mathcal{H}(N)$ be the hyperplanes of N and let

$$C_f = \{\mathrm{cl}_N(T) : T \in \binom{P \setminus \{f\}}{3},\ r_N(T) = 3\}$$

be the hyperplanes spanned by triples surviving deletion of f. Let A_f be $\mathcal{R}(N \setminus f)$ compressed to C_f and let A be $\mathcal{R}(N)$ compressed to $\mathcal{H}(N)$.

- We get a compressed version of $\mathcal{S}(N)$,

$$\widehat{\mathcal{S}}(N) = \{y \in \{0,1\}^{\mathcal{H}(N)} : y|_{C_f} \in A_f \text{ for every } f \in P\} \setminus A.$$

Step 2.4

Produce the table for $\widehat{\mathcal{S}}(N)$. We want to combine the requirements A_f for all $f \in P$ and remove the rows in A. This can be done in two ways.

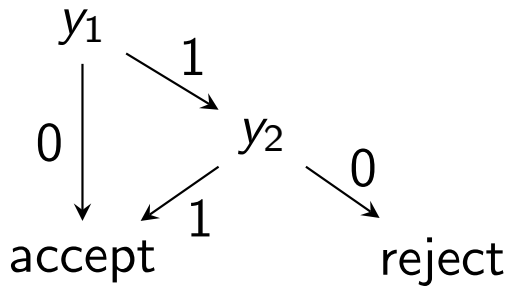
Natural join

- This is analogous to a database operation.
- A row of A_{f_1} and a row of A_{f_2} combine into a single assignment precisely when they agree on $C_{f_1} \cap C_{f_2}$.
- Build the full table for all columns in $\mathcal{H}(N)$ by joining the tables A_f one by one.
- Then remove rows that also appear in A.

An alternative is to use SAT. Use one Boolean variable y_H for each column in $\mathcal{H}(N)$.

1. Encode each A_f as a binary decision diagram and require its root to accept.
2. Block each row of A.
3. Record and block each solution until UNSAT.

Example table $\{00, 01, 11\}$



At a node testing y_H, let v be its value and ℓ, h the values of its $0, 1$ branches. Its equivalence and four clauses are

$$v \leftrightarrow ((\neg y_H \wedge \ell) \vee (y_H \wedge h))$$

$$(\neg v \vee y_H \vee \ell), \qquad (\neg v \vee \neg y_H \vee h),$$
$$(v \vee y_H \vee \neg \ell), \qquad (v \vee \neg y_H \vee \neg h).$$

To block a full row a, add $\bigvee_{H \in \mathcal{H}(N)} (y_H \neq a_H)$.

Step 3

Check which elements of $\mathcal{D}(N)$ have all single-element contractions quinary.

- Every single-element contraction of a simple rank-4 matroid has rank 3, and it suffices to check the simplifications of these contractions.
- Rank-3 simple quinary matroids can be enumerated as in Step 1. Check each simplification against the resulting catalogue.
- The members of $\mathcal{D}(N)$ with all these contractions quinary are the excluded minors. We deduplicate them up to isomorphism using Sage.

Future work

We plan to complete the classification on 11 elements across all ranks. By duality and existing work, it suffices to settle rank 5.